MSOD: A Large-Scale Multiscene Dataset and a Novel Diagonal-Geometry Loss for SAR Object Detection

Chao Wang, *Student Member, IEEE*, Wenxuan Fang, Xiang Li, Jian Yang, and Lei Luo

***Abstract*—Synthetic aperture radar (SAR) has attracted significant attention due to its excellent all-weather imaging capabilities. However, SAR image object detection methods face two major challenges: 1) most existing datasets are small in volume and single in category and scene and 2) existing intersection over union (IoU)-based loss functions cannot fully capture the relationship between prediction and target bounding boxes. To further advance the development of the SAR object detection method, we construct a large-scale multiscene SAR object detection (MSOD) dataset. It comprises three distinct scenarios, containing 40K images and about 1M instances of interest classified into six categories. In addition, we propose a novel diagonal-based similarity loss, diagonal-geometry IoU (DGIoU), to optimize the performance of SAR object detection by measuring the similarity between the diagonal of the prediction and target boxes. Specifically, we equivalently represent a rectangular box as a diagonal and then define DGIoU based on the similarity of a set of sampling points between the diagonals of the predicted box and the target box. DGIoU effectively characterizes the difference between the predicted box and the target box, particularly in box inclusion and separation cases, resulting in improved localization accuracy. Numerous experimental results demonstrate that MSOD is closer to practical application and more challenging than existing SAR image datasets and serves as a strong benchmark for evaluating the effectiveness of various IoU loss functions. The dataset and code are available at https://github.com/wchao0601/MSOD-DGIoU**

***Index Terms*—Benchmark dataset, bounding box regression, object detection, remote sensing, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has become an increasingly crucial technology in remote sensing object detection due to its exceptional all-weather imaging capabilities. Unlike traditional optical sensors, SAR can acquire imagery irrespective of conditions such as snow, cloud cover, or night, as illustrated in Fig. 1(a). Due to these remarkable advantages, SAR is widely employed in disaster exploration [1], geological prospecting [2], and object detection [3], [4]. However, despite its expanding applications, the field still faces significant challenges, such as limited availability of open-source data and inadequate detection loss functions.

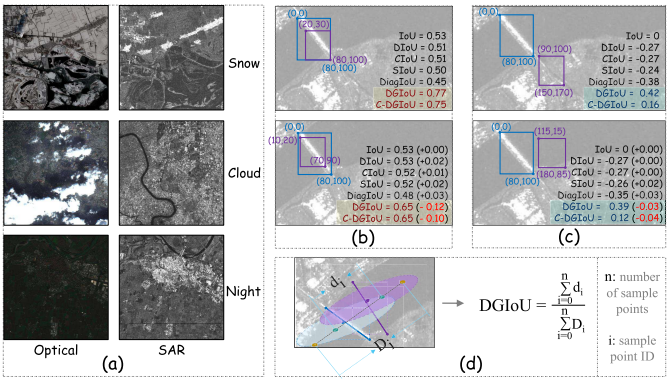


Fig. 1. (a) SAR images offer the advantage of being unaffected by conditions like snow, cloud cover, or the absence of sunlight. (b) DGIoU demonstrates greater sensitivity to object position changes in scenarios involving bounding box inclusion. (c) DGIoU exhibits greater sensitivity to object position changes in scenarios involving bounding box separation. (d) Illustration of the DGIoU loss definition used for bounding box regression.

Recently, deep learning has achieved outstanding success in optical image object detection, largely driven by the availability of large-scale, richly annotated datasets such as DIOR [5] and DOTA [6]. In contrast, research on SAR image object detection has progressed considerably more slowly. This gap is primarily due to the scarcity of high-quality, diverse, and large-scale labeled SAR datasets. Most existing SAR datasets [7], [8], [9], [10] are relatively small in scale and typically focus on a single-object category, thereby limiting their applicability in complex scenarios. For instance, the SAR-AIRcraft-1.0 dataset [11] primarily focuses on aircraft detection, while the ShipDataset [12] is mainly designed for maritime vessels. Although recent datasets such as SARDet-100K [13] and RSAR [14] have alleviated some of these limitations, they are often generated by harmonizing SAR images from multiple sources, a process that may introduce domain bias at the data input stage and compromise the fairness and generalizability of model evaluation. Furthermore, the distinctive imaging principles of SAR introduce fundamental challenges for object

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detection. Unlike optical images, SAR images exhibit a strong sensitivity to surface texture and geometric features, leading to significant variations in object appearance across different viewing angles and polarization modes. These variations often cause blurring or deformation of target boundaries, making manual annotation both labor-intensive and error-prone. Moreover, interpreting SAR images typically requires specialized domain knowledge, making it difficult to acquire large-scale, high-quality annotations.

Although existing studies have made progress in broadening object categories and enlarging dataset scales, challenges such as inefficient annotation and limited scene diversity persist. To address these issues, we constructed a multiscene SAR object detection (MSOD) dataset, based on imagery from Sentinel-1 [15] and Sentinel-2 [16]. The dataset contains two polarization modes [vertical–horizontal (VH) and vertical–vertical (VV)], three diverse scenarios (snow cover, cloud occlusion, and night), six object categories, 40K images, and around 1M labeled instances. To the best of our knowledge, MSOD is the first SAR object detection dataset featuring one million annotated instances. This contribution is anticipated to facilitate continued progress in SAR object detection.

Through extensive experiments and in-depth analysis, we have uncovered another critical obstacle in SAR image object detection: the limited applicability of existing loss functions. In SAR object detection, traditional rectangular bounding box-based object representation methods struggle to accurately describe the position and shape of targets. Specifically, the imaging characteristics of SAR can cause the edges of objects to appear blurred, fragmented, or irregular, which is more complex compared to the target boundaries in optical images. Traditional rectangular bounding box methods typically assume that each object is represented by a complete and uniform rectangular region. However, this assumption often does not hold true for SAR images, affecting the accuracy of detection results. Additionally, loss functions relying on intersection over union (IoU) might fail to precisely capture the discrepancies between predicted and ground-truth bounding boxes. In SAR images, due to the blurriness and irregularity of object boundaries, IoU values may inaccurately represent the true disparity between the target and predicted boxes. While improved IoU loss functions can mitigate the problem to some extent, they still face limitations in precisely capturing object contours and positions. We compare the performance of different IoU loss functions (e.g., distance IoU (DIoU) [19], complete IoU (CIoU) [20], SIoU [21], and diagonal IoU (Diag-IoU) [22]) in bounding box inclusion and separation scenarios. The results in Fig. 1(b) and (c) demonstrate that these methods exhibit limited sensitivity to variations in object position under such scenarios.

Zheng et al. [23] demonstrated that a rectangular box can be uniquely defined by its diagonal. Therefore, we quantize the bounding box as a diagonal, which can effectively capture object geometries in the context of SAR imaging. To address the limitations of traditional IoU-based loss functions in SAR object detection, we introduce a novel diagonal geometry loss function, diagonal-geometry IoU (DGIoU), that quantifies the similarity between the diagonal of the prediction and target boxes by the similarity of the sampled points. Unlike CLRNet's y-aligned sampling strategy [23], our approach uniformly samples points based on proportional distances along the diagonal, thereby ensuring consistent alignment irrespective of object orientation. We further enhance Diag-IoU [22] by applying weighted sampling to more accurately capture geometric discrepancies between predicted and ground-truth diagonals. As illustrated in Fig. 1(b) and (c), our proposed DGIoU and its variant C-DGIoU are more sensitive to object position changes compared to existing IoU-based loss functions (see [19], [20], [21], and [22]), particularly in complex SAR scenarios.

In summary, our main contributions are as follows.

1) MSOD is a multipolarization, multiscene SAR object detection dataset containing 40 000 images and 9 98 898 labeled instances. The dataset covers a rich variety of realistic scenarios and target categories, providing important support for SAR object detection research. To the best of our knowledge, MSOD is the first SAR dataset containing nearly one million annotated instances.
2) To more accurately measure the dissimilarity between the diagonals of predicted and target bounding boxes, we introduce a novel loss function, DGIoU. In addition, we design an effective diagonal-based sampling strategy that adaptively assigns weights to individual points.

II. Related Work

A. SAR Object Detection

In recent years, object detection frameworks have shown significant performance in natural scenes, including the RCNN series (Faster-RCNN [24], Oriented R-CNN [25], RetinaNet [26], FCOS [27], GFL [28], and DETR series [29], [30]) and YOLO series [31], [32], [33], [34]. Nevertheless, SAR image object detection continues to encounter significant difficulties because of the sparse information and scattering noise intrinsic to SAR imagery. General object detection models cannot be directly applied to SAR images. To address these problems, the MSSDNet [35] enhances the network's feature extraction ability and balances ship detection across multiple scales in SAR images through the design of specialized modules. Although most existing studies concentrate on reducing SAR scattering noise interference by enhancing network architectures, there have been limited efforts to tackle this issue at the level of target representation. Consequently, [36] employs point features to represent aircraft in SAR images and introduces an incentive attention feature fusion (IAFF) strategy that emphasizes crucial semantic areas of the target through these point features.

Unlike the above approaches, DGIoU is inspired by the cross-layer refinement network (CLRNet) [23] and Diag-IoU [22] frameworks. Specifically, we equivalently represent a bounding box as a diagonal line. The diagonal line contains the geometric factors of the bounding box, and the similarity of the diagonal line can effectively represent the differences of the bounding box, a paradigm that has not yet been adequately investigated in existing studies.

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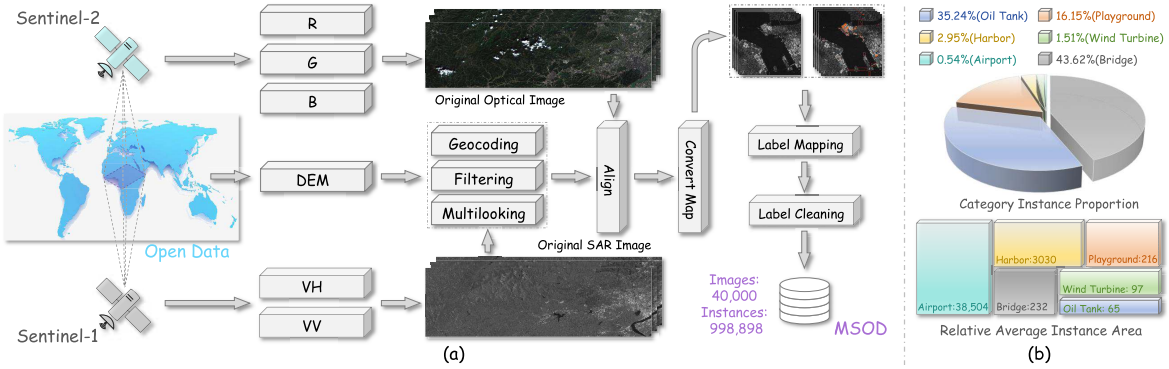


Fig. 2. (a) Construction pipeline of the MSOD dataset. (b) Proportion of instances per category and the corresponding average area (in pixels) in MSOD.

B. Bounding Box Regression Loss

Bounding box regression plays a critical role in enhancing the spatial accuracy of predicted boxes. Earlier works predominantly adopted l_n-norm loss formulations, including l_1, l_2, and smooth-l_1 losses. However, these losses ignore the intrinsic correlation of the prediction values and fail to match the IoU evaluation metrics. In SAR object detection, the IoU loss approach introduced by [37] has demonstrated notable success. As shown in Fig. 1(b), a considerable deviation between the prediction and ground-truth bounding boxes may impair the detection model's performance to varying degrees.

To address this problem, generalized IoU (GIoU) [38] adjusts the loss function by incorporating the smallest enclosing box that covers both the predicted and ground-truth bounding boxes, allowing for effective gradient calculation even when the boxes do not overlap. This adjustment improves the stability of the object detection model. DIoU [19] enhances optimization of target bounding box regression by taking into account overlap, the distance between box centers, and scale differences. CIoU [20] additionally incorporates center point and diagonal distance, thereby improving similarity estimation for targets with irregular shapes and significant size variations. Efficient IoU (EIoU) [39] builds on CIoU by separately addressing the aspect ratio components of the predicted and ground-truth boxes, which speeds up the convergence of the model.

According to Fig. 1(b), standard techniques (e.g., IoU, DIoU, CIoU, SIoU, and Diag-IoU) exhibit limitations in scenarios involving bounding box containment, primarily due to their disregard for positional and geometric characteristics embedded in the diagonal. Horizontal rectangular boxes cannot accurately describe objects with angles (e.g., bridges, airports, and playgrounds), primarily due to the presence of extensive background content. By comparison, the diagonal more precisely represents the target's location and shape in the SAR image. As depicted in Fig. 1(d), DGIoU enhances the positional estimation accuracy by offering a more refined measure of the bounding box offset.

III. Multipolarization SAR Object Detection (MSOD) Dataset

This section presents MSOD, a new multiscene SAR object detection dataset gathered from Sentinel-1 [15] and Sentinel-2 [16] satellites. Notably, it is the first dataset to exceed 1M object instances, comprising a total of 40k images. This work consists of collecting, processing, aligning, and labeling multipolarization data [Fig. 2(a)], and then manually cleaning and standardizing these datasets into a uniform format.

A. Motivation of MSOD

To construct a large-scale instance set suitable for training high-capacity deep models, we strategically curate both target scenarios and object categories when building our dataset. The primary motivations behind this design are outlined below.

1) To simulate real-world environments and capture diverse object categories, we select geographically diverse cities and coastal regions across the globe, ensuring a wide range of representative samples.
2) Given the inherent complexity of annotating SAR images, particularly for nonexperts, we utilize co-registered optical images to facilitate interpretation and substantially alleviate annotation challenges.
3) The annotation used for benchmarking is the horizontal bounding box, aligning with the standard horizontal object detection protocol for evaluation and comparison.

B. Data Collection

1) Data From Sentinel-1: Sentinel-1 [15] is equipped with C-band SAR sensors, enabling it to acquire images in any weather. Sentinel-1 offers dual-polarization capability and out-of-the-box geolocation accuracy. For the SAR images in our MSOD dataset, VH and VV data obtained in the most commonly used a single look complex (SLC) mode were used. Sentinel-1 satellites provide SAR images of different regions of the globe at different times of day, which makes them particularly beneficial for feature identification tasks.

2) Data From Sentinel-2: Sentinel-2 [16] has provided high-resolution observations of Earth's land surfaces in recent years. Under cloud-free conditions, it attains a revisit interval of up to ten days at the equator. For the optical component of MSOD, we selected tiles with cloud coverage of 10%. Only the red, green, and blue bands were used to generate visually realistic RGB composites. As Sentinel-2 data are delivered

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as orthorectified and georeferenced granules rather than raw satellite imagery, no additional geocorrection is required.

We obtained SAR images and corresponding digital elevation model (DEM) data for our region of interest from the Sentinel-1 [15] satellite, selecting images with two polarization modes: VH and VV. For optical data, we downloaded Sentinel-2 [16] images corresponding to the Sentinel-1 coverage, including the red, green, and blue channels. To minimize feature discrepancies between optical and SAR images, we chose images with a time interval of less than ten days. This short time period reduces the impact of changes in targets such as bridges and oil tanks, ensuring more consistent label transfer. Our research area includes globally significant port cities and coastal regions, featuring targets of interest such as bridges, harbors, oil tanks, playgrounds, airports, and wind turbines.

To enhance data diversity, we collected imagery from a wide range of cities and regions worldwide using data from both Sentinel-1 and Sentinel-2 satellites. The MSOD dataset collects 1014 images (338 optical images, 338 SAR images with polarization mode VH, and 338 SAR images with polarization mode VV) acquired from more than 100 cities or areas. The images were acquired from 2020 to 2022 and have an image size of $10\,980 \times 10\,980$ pixels.

C. Data Processing

The data processing is shown in Fig. 2. We process the acquired data using ENVI[1] software. First, we merge the three optical channels to create the composite optical image. For SAR data, we use SLC data and apply multilooking processing, averaging the SLC data in both azimuthal and range directions. This multilook processing reduces the spatial resolution but enhances the radiometric resolution. We then perform a single filtering step on the SAR image to reduce speckle noise. Finally, we carry out geocoding and radiometric calibration and perform geometric alignment and correction of the SAR images in conjunction with the DEM data to improve coordinate accuracy. After processing the optical and SAR data, we align them based on their geographic coordinate information. Finally, we crop the image to concentrate on the region containing the target.

D. Data Labeling

To accurately and efficiently label instances in the images, we labeled the optical images of MSOD using the roLabelme toolkit, which allows for highly zoomed-in operations, thus enabling fine-grained labeling. In terms of category selection, since the data acquisition times of the Sentinel-1 and Sentinel-2 satellites could not be fully synchronized, there could be a time bias when it came to dynamic objects. This temporal inconsistency prevents us from selecting those dynamic targets (e.g., cars, ships, etc.) whose location and state may change at different points in time. To solve this problem, we selected some static and more stable targets of interest by organizing the data and referring to related work [3], [13], [40], [41], [42]. These targets include bridges, harbors, oil tanks, playgrounds, airports, and wind turbines. This selection guarantees spatiotemporal consistency of the data and enhances the accuracy of label transfer. By aligning the optical and SAR images, annotations from the optical domain can be directly mapped to the SAR domain. We present examples (Fig. 3) of target categories and scenarios included in the MSOD dataset.

Owing to the distinctive imaging properties of SAR images, target edges are often blurred, which poses difficulties for nonexperts in accurately recognizing and annotating objects. In contrast, optical images provide clear visual information that aids in the identification and labeling of targets. In particular, Sentinel-1 and Sentinel-2 are in the same group of satellites, which are more efficient in data collection, processing, and alignment. Through fine alignment, labels on optical images can be efficiently migrated to SAR images, thus greatly reducing manual labeling costs.

E. Label Cleaning and Statistical Analysis

Since the alignment accuracy is not precise to the pixel level, noticeable deviations may occur during this process, particularly in the manual labeling stage, during which omissions and mislabeling are inevitable. To overcome this challenge, we employ an excellent detector [43] to facilitate the label refinement process. The procedure is as follows: 1) we first train an object detector on optical images using existing annotations; 2) we then perform inference on all optical images to obtain predicted bounding boxes and categories; 3) we compare these predictions with the original labels and discard those with category mismatches or IoU scores below 0.75; and 4) we manually correct the remaining labels and ultimately created the high-quality MSOD dataset.

As shown in Table I, compared with existing datasets, MSOD is one of the few large-scale datasets with millions of instances, which provides an important evaluation benchmark for subsequent research. Table II presents the annotation statistics of MSOD, while Fig. 2(b) displays the distribution across different category levels. The minimum average pixel of targets in MSOD is 65 pixels, which can effectively support the detection of typical targets such as oil tanks and bridges. Existing researches (e.g., OGSOD-1.0 [3] and SARDet-100K [13]) have demonstrated the value of Sentinel-1 images.

In addition, our proposed MSOD offers a new evaluation perspective for SAR object detection.

1) MSOD comprises a substantial portion of small objects; notably, the average fuel tank occupies merely 65 pixels, rendering the dataset especially suitable for benchmarking small-object detection performance.
2) MSOD encompasses a wide range of complex imaging conditions and geographically diverse regions, enabling a more comprehensive evaluation of model robustness from a regional and semantic standpoint.
3) MSOD contains about 1M instances, offering a more comprehensive representation of the inherent properties of SAR imagery, including large spatial coverage, high object density, and complex multitarget distributions.

[1]ENVI software, available at: https://www.nv5geospatialsoftware.com

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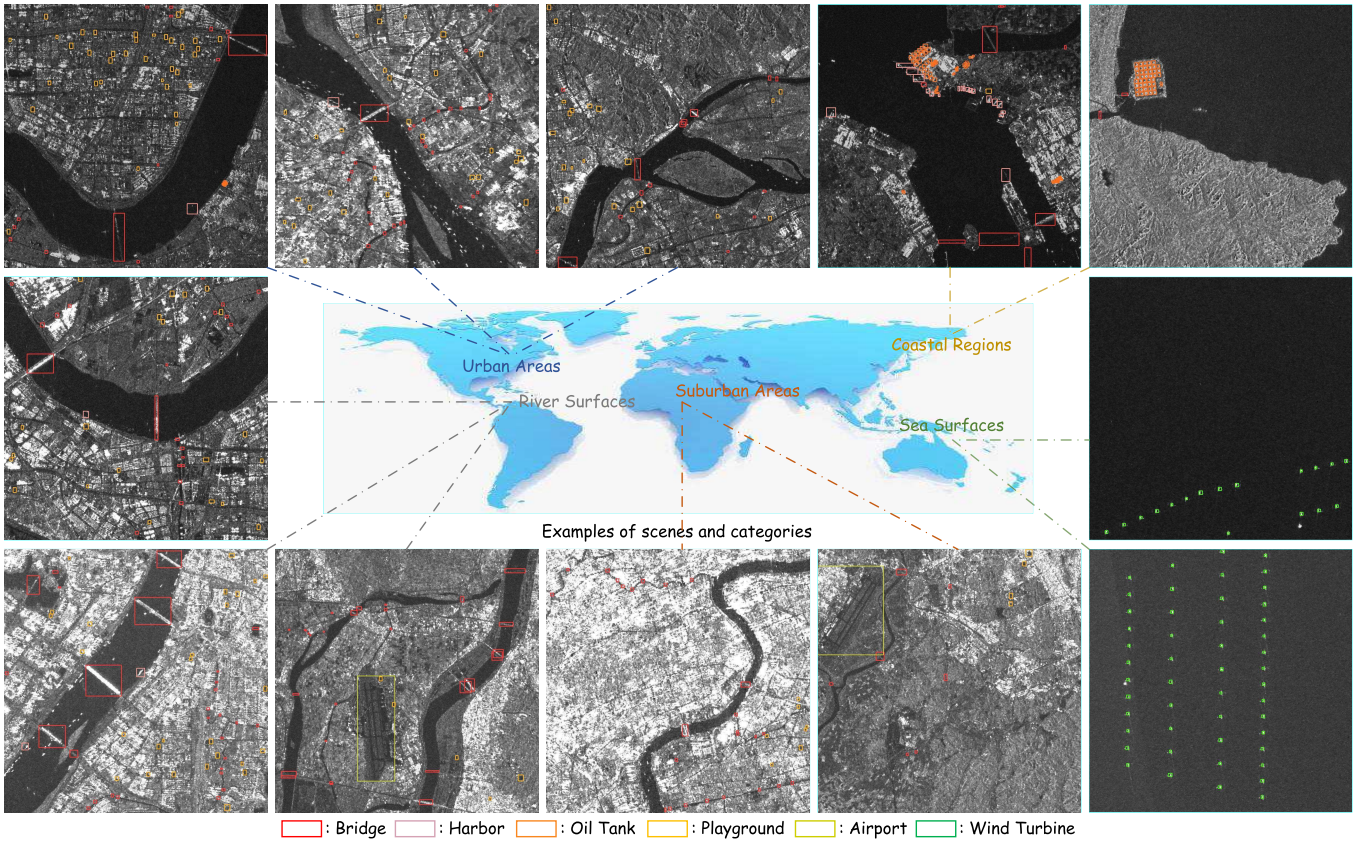


Fig. 3. Geographical distribution map of the sampled images from the proposed MSOD dataset.

TABLE I
COMPARISON BETWEEN MSOD AND OTHER DATASETS FOR SAR REMOTE SENSING IMAGES

Dataset	Year	Categories	Resolution	Polarization	Images Size	Images (Slices)	Instances	Ins/Img
ShipDataset [12]	2019	1	3m ∼ 25m	HH, HV, VH, VV	256 × 256	39,729	50,885	1.28
HRSID [7]	2020	1	0.5m ∼ 3m	HH, HV, VH, VV	800 × 800	5,604	16,969	3.03
SADD [9]	2021	1	0.5m ∼ 3m	HH	224 × 224	883	7,835	8.87
SSDD [8]	2021	1	1m ∼ 15m	HH, HV, VH, VV	512 × 512	1,160	2,587	2.23
MASR [17]	2022	4	<=1m	HH, HV, VH, VV	256 × 256	30,158	65,202	2.16
SAR-AIRcraft-1.0 [11]	2023	1	1m	Uni-polar	800 ∼ 1,500	18,888	38,475	2.04
OGSOD-1.0 [3]	2023	3	3m	VV, VH	256 × 256	18,331	48,589	2.65
SIVED [10]	2023	1	0.1m, 0.3m	VV, VH	512 × 512	1,044	12,013	11.51
SARDet-100k [13]	2024	6	0.1m ∼ 25m	HH, HV, VH, VV	512 × 512	116,598	245,653	2.11
RSAR [14]	2025	6	0.1m ∼ 25m	HH, HV, VH, VV	512 × 512	95,842	183,534	1.91
FAIR-CSAR [18]	2025	5	1m ∼ 5m	HH, HV, VH, VV	1,024 × 1,024	106,672	349,002	3.27
MSOD (Ours)	-	**6**	**3m ∼ 5m**	**VV, VH**	**1,024 × 1,024**	**40,000**	**998,898**	**24.97**

IV. METHODOLOGY

A. Analysis of IoU-Based Losses

We begin by reviewing the definition of the IoU loss

$$\mathcal{L}_{\text{IoU}} = 1 - \frac{\left|\mathbf{b}^{\text{pre}} \cap \mathbf{b}^{gt}\right|}{\left|\mathbf{b}^{\text{pre}} \cup \mathbf{b}^{gt}\right|} \tag{1}$$

where $\mathbf{b}^{\text{pre}} = (x^{\text{pre}}, y^{\text{pre}}, w^{\text{pre}}, h^{\text{pre}})$ denotes the predicted box and $\mathbf{b}^{gt} = (x^{gt}, y^{gt}, w^{gt}, h^{gt})$ is the ground-truth box. $\cap$ and $\cup$ are the intersection and union, respectively.

As is well known, when two bounding boxes do not overlap, the IoU loss remains unchanged at a constant value of 0.0. This limitation makes it challenging to accurately assess the difference between the two boxes. To mitigate the issues with IoU loss, subsequent work incorporates penalty terms for the overlap, distance, scale, and aspect ratio consistency between the two boxes, such as in GIoU [38] and CIoU [20].

However, the center distance and aspect ratio elements of the CIoU loss function become ineffective once the predicted bounding box has the same center location and aspect ratio

TABLE II
INSTANCE-LEVEL STATISTICS OF THE MSOD DATASET

Category	Train	Val	Test	Total (cls)
Bridge	216,668	85,360	133,724	435,752
Harbor	14,876	5,502	9,056	29,434
Oil Tank	178,478	68,888	104,632	351,998
Playground	79,402	30,082	51,816	161,300
Airport	2,676	1,100	1,586	5,362
Wind Turbine	7,268	2,718	5,066	15,052
Total (set)	499,368	193,650	305,880	**998,898**

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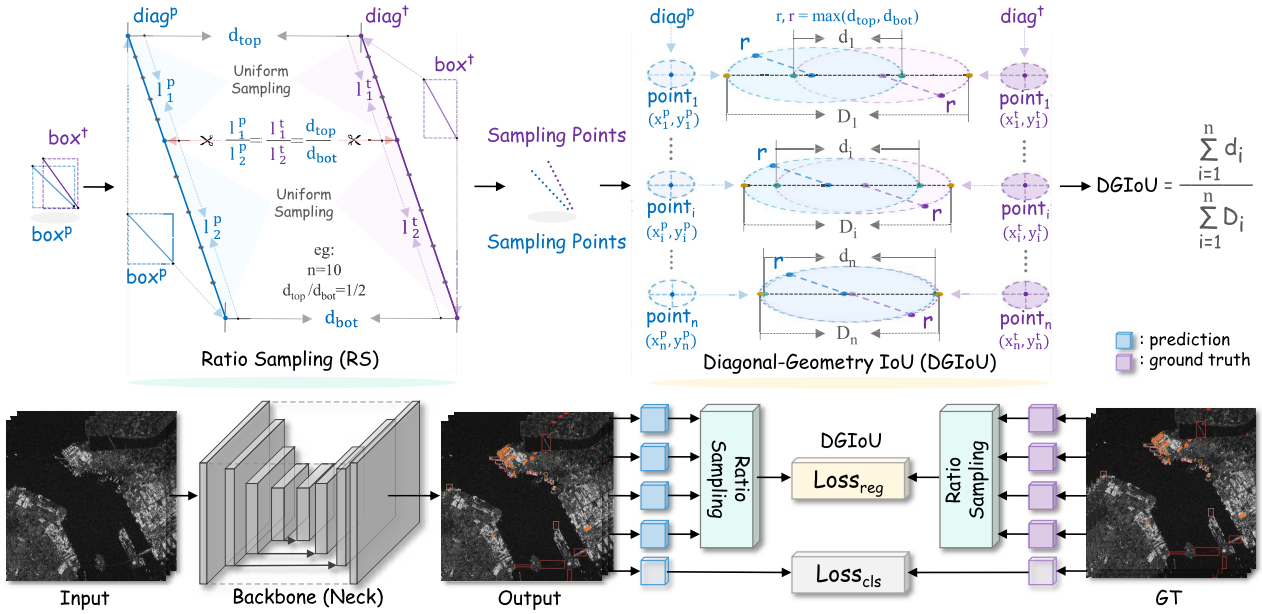


Fig. 4. Overview of the proposed DGIoU architecture. Our method incorporates a ratio-based sampling strategy that more precisely captures the relative importance of different diagonal segments, along with a structured DGIoU computation module.

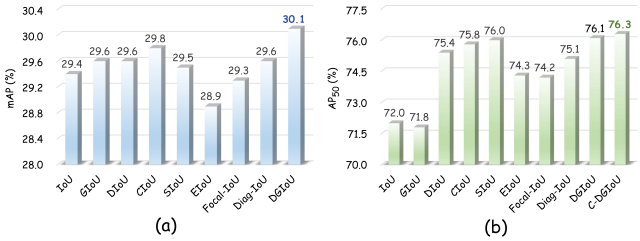


Fig. 5. (a) Comparison with popular IoU-based loss functions on the MSOD (VH) test set. (b) Comparison with popular IoU-based loss functions on the OGSOD-1.0 (SAR) test set. All results utilize the YOLOv8 detector.

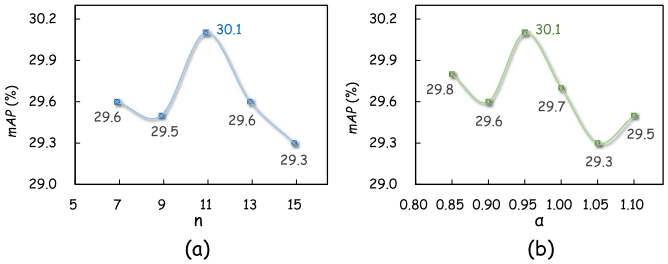


Fig. 6. (a) Impact of varying the number of sampling points on the experimental results. (b) Influence of coefficient α in the loss function. All experiments are performed on MSOD (VH) and YOLOv8 detector.

as the target box. To address this issue, Zhang et al. [22] introduced a new metric, Diag-IoU, by adding a diagonal similarity penalty term on top of IoU. The definition of Diag-IoU loss is as follows:

$$\mathcal{L}_{\text{Diag-IoU}} = \mathcal{L}_{\text{IoU}} + \lambda \mathcal{R}\left(p^{\text{pre}}, p^{gt}\right) \tag{2}$$

where p^{pre} and p^{gt} denote the sampling points on the diagonal of the prediction bounding box and the ground-truth bounding box, respectively. $\mathcal{R}(\cdot,\cdot)$ denotes the diagonal similarity and λ denotes the hyperparameters.

Although Diag-IoU achieves impressive results, the performance of Diag-IoU can be suboptimal in certain specific positions of the target and predicted boxes. Specifically, Diag-IoU does not make full use of the rich information contained in the diagonal representation. When the diagonals of the target and predicted boxes are significantly distant from each other, additional gradients are needed to accelerate loss convergence. Therefore, we reevaluated the potential of diagonals.

B. Diagonal-Geometry IoU

1) Ratio Sampling: As shown in Fig. 4, unlike the sampling method of Diag-IoU [22], we divide the two diagonals into two parts based on the ratio r_1/r_2 of the top distance d_{top} to the bottom distance d_{bot} of the two diagonals and then sample uniformly in each of these two parts. The sampling points (x_i, y_i) are calculated as

$$x_i = \begin{cases} x_1 + \dfrac{r_1}{r_1 + r_2} \times (x_n - x_1) \times \dfrac{i}{m}, & i < m \\ x_1 + \dfrac{r_1}{r_1 + r_2} \times (x_n - x_1), & i = m \\ x_m + \dfrac{r_2}{r_1 + r_2} \times (x_n - x_1) \times \dfrac{i-m}{m}, & i > m \end{cases} \tag{3}$$

$$y_i = \begin{cases} y_1 + \dfrac{r_1}{r_1 + r_2} \times (y_n - y_1) \times \dfrac{i}{m}, & i < m \\ y_1 + \dfrac{r_1}{r_1 + r_2} \times (y_n - y_1), & i = m \\ y_m + \dfrac{r_2}{r_1 + r_2} \times (y_n - y_1) \times \dfrac{i-m}{m}, & i > m \end{cases} \tag{4}$$

where (x_1, y_1) and (x_n, y_n) are the spatial locations of the top and bottom of the diagonal, respectively, and (x_m, y_m) is the dividing point of the diagonal. By experimental argument, the number of sample points n given to each diagonal is set to 11. Therefore, the diagonal is expressed as a sequence of sample points: $p = [(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)]$. The advantage of this sampling method is that more sampling points are distributed at the end, where the two diagonals are closer

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together, thus increasing the contribution weight at that end. This method more accurately reflects the importance of the diagonal components and improves the validity of the model.

2) Definition of DGIoU: As shown in Fig. 4, our proposed DGIoU method draws inspiration from the concept of spherical neurons introduced in [44]. We begin by creating two circles centered at each pair of corresponding points, using the maximum values of the top distance d_{top} and the bottom distance d_{bot} as their respective radius. Next, we link the two centers and extend them into a line segment, attaching a radius at each endpoint. Then, the point intersection d_i and union D_i between the points $p_i^p = (x_i^p, y_i^p)$ and $p_i^t = (x_i^t, y_i^t)$ are defined using the partial and extended line segments. The process of calculating d_i and D_i is as follows:

$$d_i = 2 \times \frac{\max\left(d_{\text{top}}, d_{\text{bot}}\right)}{\alpha} - \text{dis}\left(p_i^p, p_i^t\right) \tag{5}$$

$$D_i = 2 \times \frac{\max\left(d_{\text{top}}, d_{\text{bot}}\right)}{\alpha} + \text{dis}\left(p_i^p, p_i^t\right) \tag{6}$$

where α denotes the scaling factor of radius r and $\text{dis}(p_i^p, p_i^t)$ denotes the distance between p_i^p and p_i^t. In this case, the distance d_i is always positive. The distance d_i may be negative in Diag-IoU [22], thus requiring a specially designed scheme to increase the gradient to accelerate convergence.

Instead, our method avoids this problem by ensuring that the two circles constructed are always intersecting or tangent. By treating the diagonal as a whole, we evaluate the similarity between the target and predicted bounding boxes through the aggregate similarity of each paired sample point. Our proposed DGIoU loss is defined as

$$\mathcal{L}_{\text{DGIoU}} = 1 - \frac{\sum_{i=1}^{n} d_i}{\sum_{i=1}^{n} D_i}. \tag{7}$$

Unlike the $\mathcal{L}_{\text{Diag-IoU}}$ loss, our approach fully leverages the diagonal information by employing DGIoU directly as a loss function rather than as a penalty term for IoU. This is grounded in the principle that a rectangular box can be equivalently described by its diagonal. Therefore, the diagonals of bounding boxes offer comprehensive information for assessing the similarity between any two boxes, whether they overlap, are contained within one another, or are separate. As an innovative and effective alternative to IoU loss, its advantages include the following.

1) DGIoU is completely defined by the diagonal that links the predicted bounding box to the ground truth box.
2) The DGIoU treats the diagonal as a whole unit, offering comprehensive and precise guidance during training.

V. Experiments

A. Dataset and Metrics

To validate the effectiveness of DGIoU, we assess the proposed loss function on the MSOD dataset by comparing its performance with several representative object detectors. All experiments were trained on a platform with four NVIDIA RTX 4090 GPUs. The YOLO series algorithms were implemented based on Ultralytics, with the number of iterations set to 100. For example, YOLOv5 [43], YOLOv8 [43], YOLOv9 [53], and YOLOv10 [54]. We perform optimization using stochastic gradient descent (SGD), setting the initial learning rate to 0.01, momentum to 0.937, and batch size to 8. All other algorithms were implemented using the MMDetection toolkit (e.g., the FCOS [27], ATSS [45], AutoAssign [46], PAA [47], GFL [48], TOOD [49], VFNet [50], CenterNet [51], and LSKNet [52]). For MMDetection's algorithm, most models were trained for a 6× schedule that comprises a budget of 72 epochs. For all baseline detectors, we utilized the SGD optimizer with a learning rate of 0.01, momentum of 0.9, weight decay of 0.0001, and a batch size of 4. Evaluation of all models was conducted using two object detection metrics: mean average precision (mAP) and AP_{50}. The AP_{50} metric calculates the average precision across all categories at an IoU threshold of 0.50, while mAP represents the average precision averaged over IoU thresholds ranging from 0.50 to 0.95 in increments of 0.05.

B. Class Imbalance Handling

The proposed MSOD dataset exhibits substantial disparities in sample counts across classes, which biases the model toward majority classes and diminishes its ability to recognize minority classes. To address this issue, we adopt focal loss [48] as the default configuration. This loss function mitigates class imbalance by downweighting easily classified samples, thereby directing the model's attention toward harder examples and improving overall performance.

C. Main Results

Table III summarizes the experimental outcomes of different detection frameworks evaluated on the MSOD dataset. The comparisons clearly demonstrate the superiority of DGIoU. Table IV reports the comparison with other methods (e.g., DETR-based method RT-DETR [55], two-stage methods such as Faster R-CNN [56], Mask R-CNN [57], and Grid R-CNN [58], one-stage methods such as FCOS [27], YOLOv8 [43], YOLOv10 [54], and YOLOv11 [43]) on the MSOD dataset. All methods show a significant improvement in their performance when incorporating DGIoU losses. When combined with YOLOv11, we improved its detection performance by 3.3% on the AP_{50} metric (VV). In particular, the most significant boosts were taken on targets such as playgrounds and the harbor.

To further validate the effectiveness of DGIoU loss, we experimented with different losses (e.g., Focal-IoU, Diag-IoU, and our proposed DGIoU) based on the framework of YOLOv8. The results are shown in Fig. 5(a), DGIoU outperforms other excellent IoU loss functions, achieving the best results. We also evaluated the performance of DGIoU with its corresponding variants (e.g., C-DGIoU) on the OGSOD-1.0 [3] dataset. The hyperparameters (e.g., data augmentation and initial learning rate) were consistent with those used in the MSOD dataset. As shown in Fig. 5(b), C-DGIoU significantly outperforms existing CIoU methods and achieves the highest performance compared to other approaches. In particular, it

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TABLE III

PERFORMANCE COMPARISON ON THE MSOD (VH, VV) TEST SET. IN *l*|*r*, *l* DENOTES THE DETECTION RESULT IN MSOD WITH VH POLARIZATION AND *r* DENOTES THE DETECTION RESULT WITH VV POLARIZATION. T-TIME DENOTES TRAINING TIME. **BLACK** INDICATES THE BEST RESULT

Method	**T-Time (h)**		**Bridge**		**Harbor**		**Oil-Tank**		**Playground**		**Airport**		**Wind-Turbine**		**AP_{50} ↑**		**mAP ↑**	
(Name)	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV
FCOS [27]	16.3	16.4	30.9	32.8	77.2	76.6	27.0	29.4	47.2	47.8	79.9	80.2	51.1	45.7	52.2	52.1	29.1	28.9
ATSS [45]	14.0	14.5	28.6	32.5	70.8	75.2	25.6	**29.8**	46.0	48.4	78.8	80.1	45.0	44.3	49.1	51.7	27.7	29.2
AutoAssign [46]	16.8	16.8	27.1	27.6	68.5	69.0	28.7	28.3	41.0	40.7	76.3	79.7	41.5	37.3	47.2	47.1	25.1	25.1
PAA [47]	9.93	9.37	31.2	30.0	73.3	73.4	15.3	15.6	46.1	47.9	80.2	82.2	51.8	43.5	49.7	48.8	28.8	28.3
GFL [48]	17.9	17.9	28.5	31.4	71.4	71.3	20.7	25.3	44.2	45.2	79.6	80.4	44.2	42.1	48.1	48.1	28.6	29.6
TOOD [49]	17.7	17.9	29.5	32.0	75.8	76.0	28.1	27.2	43.6	44.6	78.8	80.5	47.9	41.4	50.6	50.3	28.1	28.5
VFNet [50]	19.1	19.5	31.7	31.9	75.9	77.3	26.3	26.0	47.9	45.6	80.4	85.5	44.8	43.8	51.2	51.7	29.4	29.3
CenterNet [51]	16.8	16.3	30.1	32.0	74.8	75.2	**29.1**	29.0	45.0	46.2	78.1	80.1	43.8	40.8	50.1	50.5	27.6	27.7
LSKNet [52]	18.5	18.9	29.7	31.5	78.3	76.3	19.9	19.9	50.3	50.7	88.4	87.6	**55.6**	**49.7**	53.7	52.6	29.9	29.4
YOLOv5 [43]	3.92	3.88	27.2	32.3	80.1	79.6	22.0	23.6	41.1	41.6	91.0	92.9	45.3	42.9	51.1	52.2	28.0	28.5
YOLOv8 [43]	3.24	3.25	30.3	34.5	81.2	78.3	23.1	22.6	48.2	47.4	90.5	89.3	50.9	49.0	54.1	53.5	29.8	28.4
YOLOv9 [53]	4.93	4.80	30.8	35.5	79.7	79.6	22.7	22.4	48.3	49.0	88.7	91.5	45.7	49.1	52.7	54.5	28.0	29.2
DGIoU (Ours)	3.28	3.28	**33.0**	**37.2**	**81.9**	**82.0**	24.6	23.7	**52.0**	**55.0**	**91.2**	**93.4**	52.3	48.9	**55.8**	**56.7**	**30.1**	**30.1**

TABLE IV

PERFORMANCE IMPACT OF SUBSTITUTING THE ORIGINAL LOSS WITH OUR DGIOU LOSS ACROSS VARIOUS STATE-OF-THE-ART DETECTORS. † DENOTES THE USE OF DGIOU INSTEAD OF THE ORIGINAL LOSS. **BLACK** AND MARK INDICATE THE BEST PERFORMANCE FOR THE COMPARISON

Method	**T-Time(h)**		**Bridge**		**Harbor**		**Oil-Tank**		**Playground**		**Airport**		**Wind-Turbine**		**AP_{50} ↑**		**mAP ↑**	
(Name)	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV	VH	VV
DETR − based :																		
RT-DETR-L [55]	61.3	60.8	**16.1**	20.9	56.7	65.0	8.51	10.1	17.3	26.5	79.8	71.0	31.4	38.8	35.0	38.7	14.7	15.5
RT-DETR-L †	62.0	61.8	15.0	**22.9**	**64.5**	**65.1**	**8.80**	**10.6**	**19.5**	**28.0**	**84.5**	**71.2**	**32.2**	**39.1**	**37.4**	**39.5**	**16.7**	**16.9**
Two Stage :																		
Faster R-CNN [56]	14.7	14.9	9.80	10.1	65.2	66.8	9.45	9.10	23.7	22.9	78.1	79.5	14.2	12.1	33.4	33.4	21.8	21.9
Faster R-CNN †	15.1	15.2	9.64	12.0	67.3	68.0	10.0	11.1	23.9	23.8	78.5	79.9	14.5	14.1	34.0	34.8	21.9	22.1
Mask R-CNN [57]	17.9	18.0	9.90	11.2	66.6	67.8	9.81	9.11	24.5	22.0	78.2	79.9	14.4	14.1	33.9	34.0	22.2	22.0
Mask R-CNN †	18.4	18.5	**10.8**	**11.3**	**67.3**	**69.0**	**11.1**	**10.8**	**25.5**	**23.2**	**78.9**	**80.9**	**14.5**	**13.9**	**34.7**	**34.9**	**22.5**	**22.4**
Grid R-CNN [58]	19.9	20.1	9.10	12.2	70.2	69.6	9.10	9.11	24.0	23.0	78.5	79.3	14.7	13.1	34.3	34.4	23.5	22.5
Grid R-CNN †	20.1	20.3	10.1	13.0	70.3	70.0	9.13	9.12	24.5	23.8	79.2	82.9	15.5	13.5	34.8	35.4	23.9	22.9
One Stage :																		
FCOS-R50 [27]	14.2	14.3	30.9	32.8	77.2	76.6	27.0	**29.4**	47.2	47.8	**79.9**	80.2	51.1	45.7	52.2	52.1	29.1	28.9
FCOS-R50 †	14.8	14.9	**31.2**	**32.9**	**78.2**	**77.9**	**27.5**	28.4	**48.1**	**48.8**	78.8	**81.1**	**51.7**	**47.7**	**52.6**	**52.8**	**29.8**	**29.7**
YOLOv8-S [43]	3.24	3.25	30.3	34.5	81.2	78.3	23.1	22.6	48.2	47.4	90.5	89.3	50.9	49.0	54.1	53.5	29.8	28.4
YOLOv8-S †	3.28	3.28	33.0	37.2	81.9	82.0	24.6	23.7	52.0	55.0	91.2	93.4	52.3	48.9	55.8	56.7	30.1	30.1
YOLOv10-S [54]	5.01	4.98	30.3	35.8	77.9	76.8	**22.6**	**23.6**	46.5	47.2	85.8	88.2	**50.9**	**50.4**	52.3	53.7	28.4	28.4
YOLOv10-S †	5.10	5.08	**33.5**	**39.1**	**79.0**	**78.9**	21.5	23.5	**53.5**	**54.9**	**87.7**	**88.8**	42.2	44.7	**52.9**	**55.0**	**29.0**	**28.8**
YOLOv11-S [43]	3.45	3.46	28.9	34.3	79.8	79.3	20.8	22.8	44.3	47.3	89.2	91.3	51.9	49.5	52.5	54.1	28.0	28.5
YOLOv11-S †	3.50	3.51	32.5	37.9	81.3	81.2	21.6	23.2	54.9	56.6	90.0	92.0	49.0	53.9	54.9	57.4	28.4	29.6
YOLOv12-S [59]	5.64	5.68	26.5	31.2	79.6	79.2	18.0	**20.0**	36.7	36.6	92.6	**93.4**	**50.9**	46.9	50.7	51.2	27.3	27.6
YOLOv12-S †	5.89	5.95	**37.0**	**32.4**	**80.5**	**80.9**	**20.4**	19.6	**52.3**	**52.0**	**94.8**	92.4	47.8	**52.0**	**55.5**	**55.2**	**29.4**	**29.1**

TABLE V

COMPUTATION OVERHEAD COMPARISON OF DIFFERENT IOU VARIANTS

Method	**IoU**	**GIoU**	**DIoU**	**CIoU**	**SIoU**	**EIoU**	**Diag-IoU**	**DGIoU**
FLOPs (M)	3.58	3.58	3.58	7.30	8.83	5.63	17.79	18.23
Times (ms)	1.63	1.82	2.26	3.08	5.50	2.51	12.74	13.15

improves the AP_{50} by 1.2% relative to the same type of method, Diag-IoU. Our method attains top performance across different datasets, highlighting its strong generalizability.

D. Computation Overhead Analysis

DGIoU imposes a slight computational overhead, but it is almost negligible compared to the overall model. As shown in Table IV, training time increases by just 0.7 h (≈ 1.1%) for RT-DETR and 0.05 h (≈ 1.4%) for YOLOv11.

Table V presents a comparative analysis of the computational overhead introduced by different IoU variants. For greater accuracy, we averaged the results over 10 000 runs, with a total of 64 000 input bounding box pairs. While traditional losses such as IoU, GIoU, and DIoU incur minimal computational cost (approximately 3.58M FLOPs and 1.63–2.26 ms), advanced variants like CIoU, SIoU, and EIoU show moderate increases in both FLOPs and computational time. Our proposed DGIoU introduces a slightly higher computational cost (18.23M FLOPs and 13.15 ms). However, this increase is comparable to Diag-IoU and remains in a reasonable range for real-time applications. More importantly, DGIoU is only employed for computing the bounding box regression loss during the training phase. The original IoU-based configuration is retained during inference, ensuring that DGIoU does not impact the inference speed.

E. Ablation Study

1) Effects of RS: The experimental findings presented in Table VI demonstrate that ratio sampling (RS) improves mAP by 0.8% relative to the uniform sampling baseline on the MSOD dataset. RS also achieved 76.3% on AP_{50} and 48.6% on mAP on the OGSOD-1.0 dataset, both of which are more accurate than Uni-S and Rand-S. These experimental results substantiate the validity and effectiveness of the proposed RS.

2) Effect of Different Point Numbers n on Performance: Subsequently, we discussed the effect of different sampling points on DGIoU. Fig. 6(a) displays the results,

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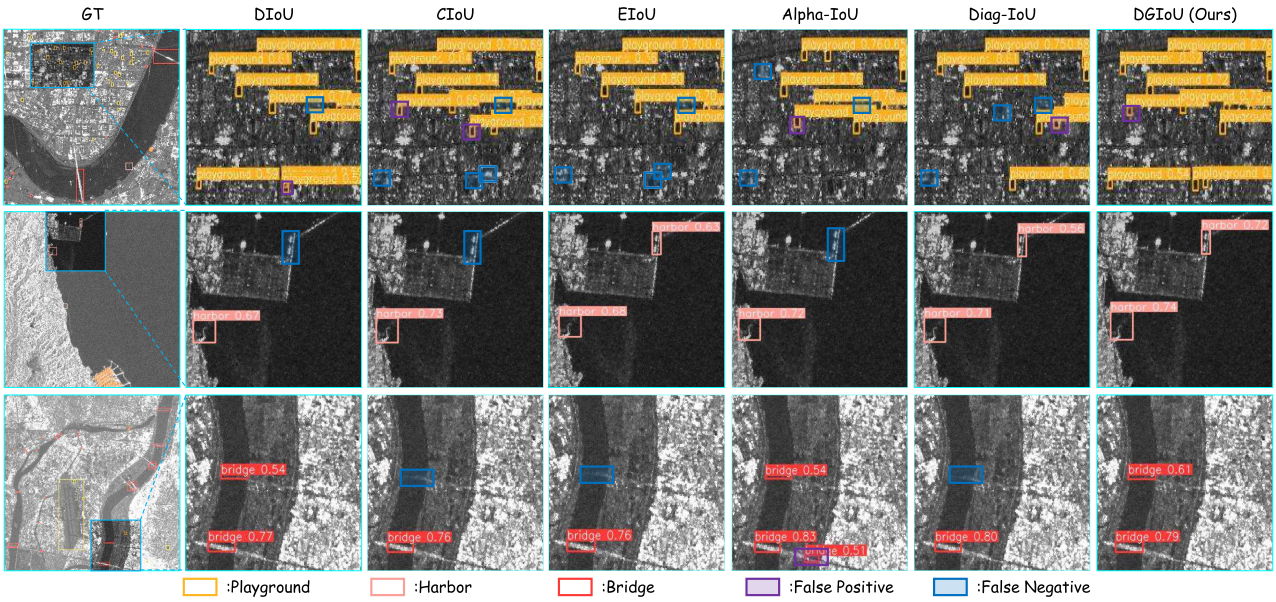


Fig. 7. Results of detection on the MSOD (VH) with our DGIoU loss and other losses. GT denotes ground truth. Results are based on the YOLOv8 detector.

TABLE VI

COMPARISON RESULTS OF DIFFERENT SAMPLING METHODS, WHERE UNI-S, RAND-S, AND RS DENOTE UNIFORM SAMPLING, RANDOM SAMPLING, AND RATIO SAMPLING, RESPECTIVELY. RESULTS ARE BASED ON THE YOLOV8 MODEL

Dataset	Method	AP_{50} ↑	mAP ↑
MSOD (VH)	Uni-S	55.1	29.3
	Rand-S	55.3	29.7
	RS (Ours)	55.8 (+**0.7**)	30.1 (+**0.8**)
OGSOD-1.0 (SAR)	Uni-S	75.8	48.1
	Rand-S	75.9	48.3
	RS (Ours)	76.3 (+**0.5**)	48.6 (+**0.5**)

showing that setting $n = 11$ yields the best performance with a 30.1% mAP. Performance declines when using too many ($n = 15$, 29.3% mAP) or too few ($n = 7$, 29.6% mAP) sampling points. This also indicates that sampling is a key factor affecting SAR object detection performance.

3) Effect of Different α on Performance: Additionally, we examined the impact of the hyperparameter α, which controls the radius length of the two circles in our proposed DGIoU. Asillustrated in Fig. 6(b), adjusting α enhances the mAP by 0.4% compared to $\alpha = 1.00$. Performance declines when α drops below 0.95, before gradually improving again. We set α to 0.95 because this value offers better control over the radius length of the constructed circles, ensuring that the two circles consistently intersect.

4) Visualization of Detection Results and Heatmaps: Fig. 7 visualizes detection results from various methods on our MSOD (VH) dataset. It clearly shows that our proposed DGIoU loss achieves more accurate detections and reduces missed targets compared to Diag-IoU and other loss functions. We randomly selected some images from the MSOD dataset and visualized $P5$ heatmaps for different IoU methods. As shown in Fig. 8, these visualization examples show that DGIoU focuses more on the target in some regions and achieves comparable performance to the best method in other regions compared to the IoU-based variants of the method. This highlights the advantages of our proposed DGIoU in improving object detection performance.

5) Generalization Capabilities of the MSOD Dataset: To further validate the generalization capability of the MSOD dataset, we designed cross-testing experiments between the two datasets, MSOD and SARDet-100K [13], for the three types of shared targets (oil tanks, bridges, and harbors). Specifically, we used the YOLOv8 (CIoU) model to train on each of the two datasets and tested it directly or fine-tuned it for a limited number of epochs on the other dataset. As shown in Table VII, without fine-tuning (None), both models have lower detection accuracy on the other dataset, making it difficult to directly compare their generalizability. Therefore, we further conducted experiments with only fine-tuning the Neck module under the condition of freezing the Backbone. The results show that models trained on MSOD and transferred to SARDet-100K can achieve very high detection accuracy. For example, after only 1 epoch (1e) of fine-tuning, the accuracy reached 26.4% (34.2% of the final accuracy of 77.1%). In contrast, a model trained on SARDet-100K and transferred to MSOD only achieved an accuracy of 9.81% (21.1% of the final accuracy of 46.4%). This strongly demonstrates that MSOD can provide stronger generalization capability compared to SARDet-100K. In addition, we evaluated the generalization ability of other representative methods (e.g., DIoU, SIoU, Diag-IoU, and the proposed DGIoU) on the MSOD and SARDet-100K datasets. The results show that these methods also perform poorly in terms of generalization between the two datasets. This indicates that the lack of generalization is not a problem specific to our method, but rather stems from the inherent challenges of transferring between these two datasets.

6) DGIoU Applies Different IoU Penalty Terms: As shown in Table VIII, we validate the extendability of DGIoU on the MSOD (VH) dataset. We apply the penalty terms of DIoU and CIoU, variants of IoU, to DGIoU. D-DGIoU and C-DGIoU further improved the AP_{50} by 1.6% and 1.5%, respectively. Experimental results demonstrate that incorporating these

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TABLE VII

COMPARISON OF GENERALIZATION CAPABILITIES BETWEEN THE MSOD (VH AND VV) AND SARDET-100K DATASETS (ONLY SHARED CLASSES TESTED)

Method	None	Pre-training on MSOD, Test on SARDet-100K						Final	None	Pre-training on SARDet-100K, Test on MSOD						Final
		1e	20e	40e	60e	80e	100e	AP_{50}		1e	20e	40e	60e	80e	100e	AP_{50}
CIoU	3.05	26.4	56.9	59.1	62.9	66.6	69.6	77.1	1.96	9.81	24.7	28.5	29.9	30.5	31.6	46.4
DIoU	2.85	27.6	56.0	59.6	63.1	66.7	70.0	76.5	1.88	8.16	24.6	28.9	30.2	30.6	31.0	46.1
SIoU	3.18	28.0	56.3	59.0	62.4	66.0	69.7	77.0	3.81	8.10	24.4	28.2	29.6	30.1	30.5	46.3
Diag-IoU	3.18	27.7	56.7	60.0	63.0	66.0	69.5	77.5	3.92	8.33	25.2	28.7	30.0	30.5	31.0	46.8
DGIoU	3.21	26.6	58.2	62.3	66.5	70.2	73.9	81.2	2.02	9.98	25.6	29.4	30.6	31.1	31.5	47.5

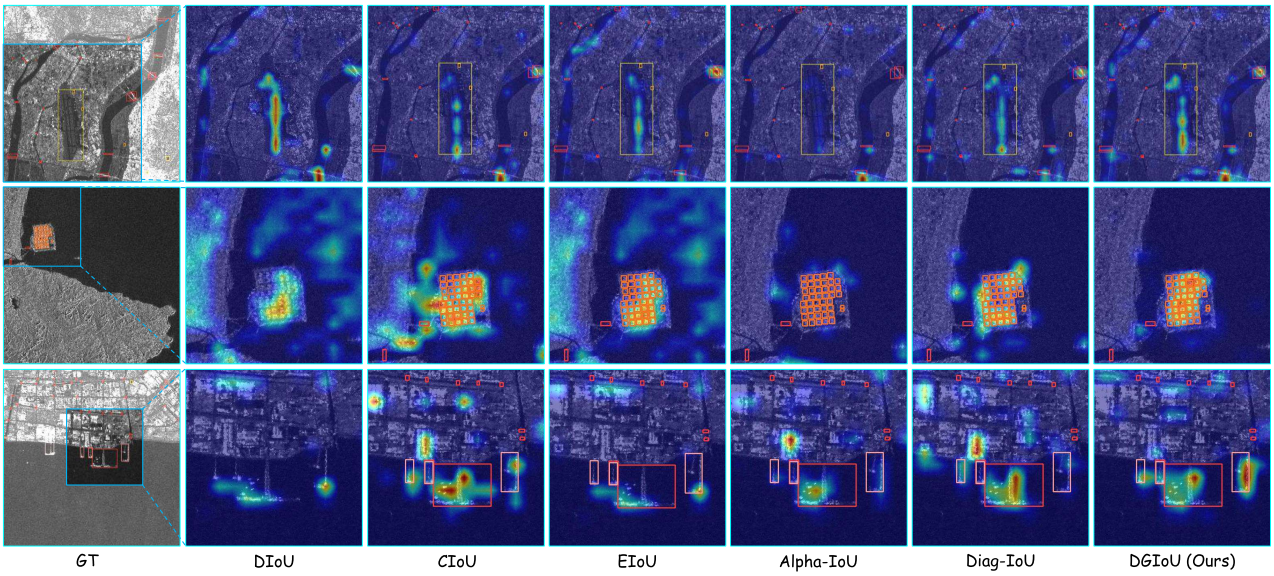


Fig. 8. Heatmap (Grad-CAM) comparison of different IoU-based loss functions. Compared with other methods, DGIoU activates the target region more intensively and effectively suppresses the irrelevant background response, indicating improved positioning accuracy.

TABLE VIII

RESULTS OF REPLACING THE STANDARD IOU TERM IN EXISTING LOSS FUNCTIONS (E.G., DIOU AND CIOU) WITH DGIOU

Method	B ↑	H ↑	O ↑	P ↑	A ↑	W ↑	AP_{50} ↑
IoU	29.7	81.0	22.7	46.9	91.2	48.0	53.2
DGIoU	30.9	81.8	19.9	53.6	91.6	47.8	54.3
D-DGIoU	32.3	**82.6**	22.8	**54.4**	**92.0**	51.4	**55.9**
C-DGIoU	**33.0**	81.9	**24.6**	52.0	91.2	**52.3**	55.8

penalty terms further enhances the method's performance. This demonstrates the effectiveness of DGIoU in replacing IoU.

F. Application of DGIoU in Oriented Object Detection

In horizontal object detection, DGIoU is implemented based on the core principle that a diagonal uniquely defines a horizontal rectangular box. So, is DGIoU equally applicable to oriented object detection? To this end, we have conducted a preliminary exploration. As shown in Fig. 9, two diagonal lines can uniquely determine an oriented rectangular box. Therefore, DGIoU has the same effectiveness in oriented object detection. The process of calculating d_i^j and D_i^j is as follows:

$$d_i^j = 2 \times \frac{\max\left(d_{\text{top}}^1, d_{\text{dot}}^1, d_{\text{top}}^2, d_{\text{dot}}^2\right)}{\alpha} - \text{dis}\left(p_i^p, p_i^t\right) \tag{8}$$

$$D_i^j = 2 \times \frac{\max\left(d_{\text{top}}^1, d_{\text{dot}}^1, d_{\text{top}}^2, d_{\text{dot}}^2\right)}{\alpha} + \text{dis}\left(p_i^p, p_i^t\right). \tag{9}$$

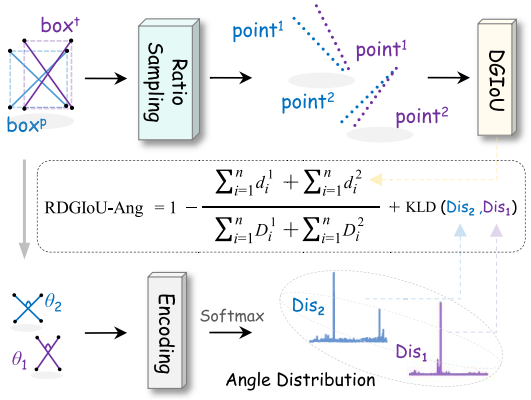


Fig. 9. Architecture of the proposed RDGIoU. RDGIoU-Ang denotes RDGIoU with angular θ distillation.

The $\mathcal{L}_{\text{RDGIoU}}$ loss is defined as

$$\mathcal{L}_{\text{RDGIoU}} = 1 - \frac{\sum_{i=1}^n d_i^1 + \sum_{i=1}^n d_i^2}{\sum_{i=1}^n D_i^1 + \sum_{i=1}^n D_i^2}. \tag{10}$$

To better exploit the potential of DGIoU in oriented object detection, we add constraints on angles to DGIoU. This is done by cosine coding the diagonal angle and using the KLD loss as a constraint. This simplified approach aims to showcase the strong potential of DGIoU in oriented object detection. Therefore, we conducted only a preliminary comparison with the

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TABLE IX

IMPACT OF DGIOU ON ORIENTED OBJECT DETECTION, EVALUATED ON THE HRSC2016 (OBB) DATASET

Method	AP_{50} ↑	mAP ↑
ProbIoU	95.5	47.8
RDGIoU (Ours)	95.1 (- 0.4)	49.1 (+1.3)
RDGIoU-Ang (Ours)	95.8 (+**0.3**)	49.4 (+**1.6**)

TABLE X

IMPACT OF DGIOU ON ORIENTED OBJECT DETECTION, EVALUATED THROUGH EXPERIMENTS CONDUCTED ON THE DIOR (OBB) DATASET

Category	ProbIoU	RDGIoU	RDGIoU-Ang
Airplane	89.4	92.4 (+**3.0**)	92.3 (+2.9)
Airport	67.3	66.6 (-0.7)	69.3 (+**2.0**)
Baseball-Field	89.6	88.7 (-0.9)	89.0 (-0.6)
Basketball-Court	93.8	93.7 (-0.1)	93.7 (-0.1)
Bridge	62.9	61.4 (-1.5)	62.0 (-0.9)
Chimney	83.6	83.9 (+0.3)	84.4 (+**0.8**)
Dam	55.5	56.5 (+**1.0**)	55.5 (+0.0)
E-Service-Area	94.4	93.9 (-0.5)	94.4 (+0.0)
E-Toll-Station	86.1	85.3 (-0.8)	85.1 (-1.0)
Golf-Field	85.4	87.0 (+**2.6**)	86.9 (+1.5)
Ground Track Field	81.0	84.9 (+**3.9**)	84.3 (+3.3)
Harbor	65.0	64.8 (-0.2)	64.7 (-0.3)
Overpass	71.8	70.6 (-1.2)	71.7 (-0.1)
Ship	95.2	95.3 (+**0.1**)	95.0 (-0.2)
Stadium	78.4	81.9 (+**3.5**)	80.9 (+2.5)
Storage-Tank	86.4	87.0 (+0.6)	87.4 (+**1.0**)
Tennis-Court	94.5	94.5 (+0.0)	94.6 (+**0.1**)
Trainstation	76.7	74.4 (-2.3)	76.4 (-0.3)
Vehicle	67.5	65.5 (-2.0)	65.6 (-1.9)
Windmill	92.0	91.5 (-0.5)	91.7 (-0.3)
AP_{50} ↑	80.8	81.0 (+0.2)	81.3 (+**0.5**)

YOLOv8 implementation of ProbIoU [43]. The $\mathcal{L}_{\text{RDGIoU-Ang}}$ loss is defined as

$$\mathcal{L}_{\text{RDGIoU-Ang}} = \mathcal{L}_{\text{RDGIoU}} + \mathcal{L}_{\text{KLD}}\left(\mathcal{S}\left(\mathcal{C}\left(\theta_1\right)\right), \mathcal{S}\left(\mathcal{C}\left(\theta_2\right)\right)\right) \quad (11)$$

where $\mathcal{L}_{\text{KLD}}$ denotes the KL scattering loss, $\mathcal{S}(\cdot)$ denotes Softmax, and $\mathcal{C}(\cdot)$ denotes the cosine code. θ_1 and θ_2 represent the angles formed by the diagonals of the predicted and the ground-truth box, respectively.

1) Experimental Results on HRSC2016: The HRSC2016 [60] dataset, publicly released in 2016, has become a widely used benchmark in remote sensing images. The dataset suffers from challenging problems such as high overlap of labeled frames, complex backgrounds, and high similarity between ships to be detected and nearshore textures. HRSC2016 contains 1061 images and 2976 instances. In our experiments, we uniformly scaled the size of all images to 800×800.

As shown in Table IX, RDGIoU achieves an accuracy of 95.1% on AP_{50}, which is 0.4% lower compared to the baseline ProbIoU. However, it is 1.3% higher on mAP, which demonstrates that DGIoU can provide a more accurate target box. RDGIoU-Ang with added angular constraints is 1.6% higher than ProbIoU, which demonstrates that DGIoU has further potential to be explored, and the angular constraints are only the result of our initial exploration.

2) Experimental Results on DIOR: DIOR [5] includes 5862 training images, 5863 validation images, and 11 725 testing images. It covers 20 object categories, with each image having a resolution of 800×800 pixels. We combine the training and validation sets for model training and assess the performance of DGIoU using the official test set.

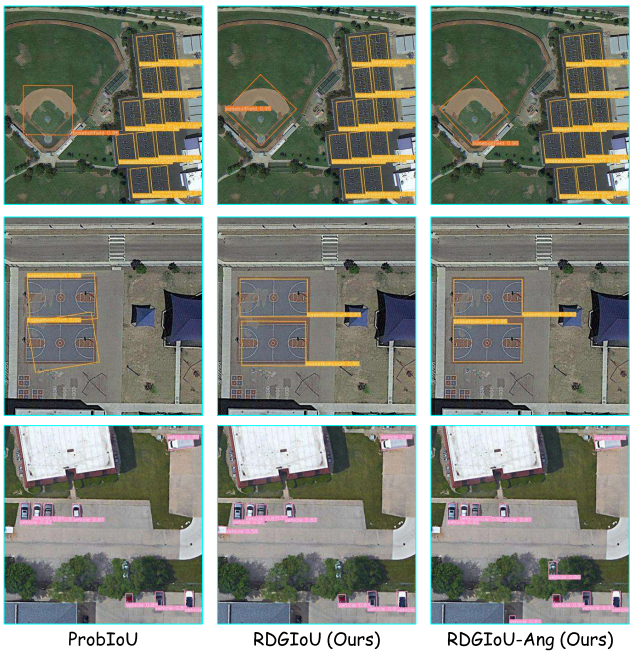


Fig. 10. Detection results using DGIoU on the DIOR dataset.

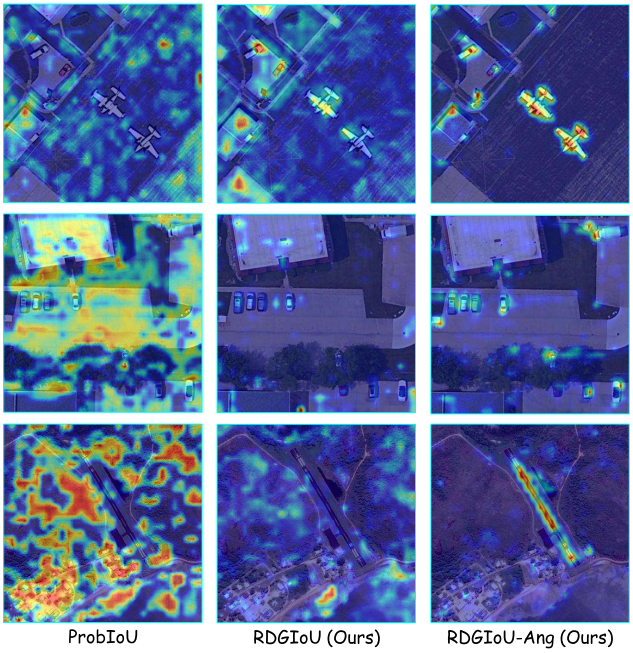


Fig. 11. Heatmap visualization comparison on the DIOR dataset.

As shown in Table X, the results indicate that the proposed DGIoU achieves a comparable AP_{50} performance to the ProbIoU method on the DIOR dataset. Moreover, RDGIoU-Ang, which incorporates additional processing of angular information, attains the highest mAP of 81.3%. These results show that RDGIoU-Ang also exhibits great competitiveness in the oriented object detection task.

3) Visualization Results: Fig. 10 visualizes a set of representative detection results, clearly demonstrating the efficacy of the proposed model in achieving accurate target localization. Our method can correctly detect some challenging

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invisible objects, such as vehicles, and provide more precise angles for some objects, such as targets like basketball fields and baseball fields.

4) Heatmap Results: Random samples from the DIOR dataset were used to visualize and compare the $P5$ heatmaps produced by different models. As illustrated in Fig. 11, the visualizations demonstrate that our DGIoU approach exhibits a stronger focus on target regions compared to ProbIoU. Furthermore, the incorporation of angle constraints enables DGIoU to achieve more precise localization and enhanced detection performance. This demonstrates that DGIoU has some potential to be explored for oriented object detection.

VI. Conclusion and Future Work

In this article, we present a new multiscene dataset for SAR object detection, MSOD, along with a novel diagonal-based bounding box regression loss, DGIoU. The MSOD dataset comprises over 40 000 SAR images and approximately one million annotated instances, offering a rich and diverse resource for advancing SAR object detection research. We present DGIoU to address the performance degradation of the existing IoU-based loss in SAR images. This newly designed loss captures the spatial correspondence between predictions and ground truth more effectively by leveraging diagonal geometry, resulting in improved localization accuracy and robustness. In future work, we intend to examine the potential of DGIoU in handling oriented object detection tasks.

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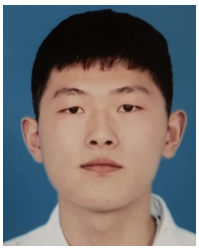
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